



دانشگاه علوم پزشکی تهران

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بلوک تنفس

گروه الف

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۵

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- مشخصات دانشجویی خود را روی صفحه اول پرسشنامه بنویسید.
- لطفاً سوالات پرسشنامه را به دقت مطالعه، سپس گزینه صحیح را در پاسخنامه با مداد مشکی وارد نمایید.
- تعداد ۲۹ سؤال تستی در ۵ صفحه می باشد.

Anatomy

1. All of the following bones are involved in the formation of the posterior nasal aperture (choana) except...?

- a) Vomer
- b) Palatine
- c) Sphenoid
- d) Maxilla

2. Which of the following arteries branches from the maxillary artery, which is a branch of the external carotid artery?

- a) Sphenopalatine
- b) Anterior Ethmoidal
- c) Posterior Ethmoidal
- d) Superior labial

3. Which of the following options is correct?

- a) The ethmoidal bulla is located below the inferior concha.
- b) The maxillary sinus opens into the Semilunar hiatus.
- c) The nasolacrimal duct opens into the middle meatus.
- d) The middle ethmoidal air cells open into the superior meatus.

4. All of the following structures are adjacent to the medial surface of the right lung except...?

- a) Arch of azygos vein
- b) Thoracic duct
- c) Superior vena cava
- d) Inferior vena cava

5. Which of the following options is a characteristic of the parietal pleura?

- a) It is innervated by the sympathetic nerve.
- b) It penetrates into the deep fissures.
- c) The inferior border of the parietal pleura at the midclavicular line is at the level of the eighth rib.
- d) The pulmonary ligament is derived from the visceral pleura.

6. All of the following options are correct except...?

- a) The right main bronchus is wider than the left main bronchus.
- b) The right main bronchus is shorter than the left main bronchus.
- c) The left main bronchus is further from the midline compared to the right main bronchus.
- d) The left main bronchus divides before entering the hilum of the right lung.

7. Which of the following options is incorrect?

- a) The upper border of the quadrangular membrane forms the aryepiglottic fold.
- b) The aryepiglottic fold plays a role in the formation of the laryngeal inlet.
- c) The false vocal fold is attached anteriorly to the arytenoid cartilage.
- d) The upper border of the cricothyroid membrane (conus elasticus) forms the true vocal cords.

8. All of the following are correct about the internal laryngeal nerve, except:

- a) It pierces the thyrohyoid membrane.
- b) It has muscular branches.
- c) It supplies sensation to the laryngeal mucosa above the true vocal cords.
- d) This nerve is one of the two terminal branches of the superior laryngeal nerve.

Embryology

9. The lung bud originates from which of the following parts?

- a) Endoderm
- b) Visceral mesoderm
- c) From the sixth pharyngeal arch
- d) Foregut

Histology

10. In the structure of the olfactory epithelium, which of the following is correct?

- a) It contains bipolar sensory neurons.
- b) Bowman's serous glands cause lubrication of the duct.
- c) The axons of olfactory neurons are in the lamina propria.
- d) Olfactory cells have basal cilia.

11. In the structure of the respiratory epithelium, which of the following is correct?

- a) Kulchitsky cells have apical granules.
- b) Brush cells have apical microvilli.
- c) Goblet cells contain lipoprotein.
- d) Columnar cells have apical nuclei.

12. In the histological structure of the trachea:

- a) The epithelial cells lack granules.
- b) The C-shaped hyaline cartilage is under the adventitia.
- c) Mucous glands in the submucosa produce a watery serous secretion.
- d) The ends of the tracheal cartilage are covered by the trachealis skeletal muscle.

13. Regarding the histological structure of the lung, which of the following is correct?

- a) The respiratory bronchiole has cartilaginous plates.
- b) The terminal bronchiole has seromucous glands.
- c) In the terminal bronchiole, the height of the epithelium is greater.
- d) Clara cells appear in the terminal bronchiole.

Physiology

14. If in a patient with pneumonia the diffusing capacity of CO₂ is 300 ml/min/mmHg, what will be the diffusing capacity of oxygen in this person (in ml/min/mmHg)?

- a) 12.5
- b) 15
- c) 20
- d) 30

15. In which generation of the lower airways is the resistance highest?

- a) Left and right main bronchi
- b) Tenth generation bronchi
- c) Fifteenth generation terminal bronchioles
- d) The last generation of respiratory bronchioles

16. What is the difference in intra-alveolar pressure (in cm H₂O) at the end of a normal inspiration compared to the beginning?

- a) -1
- b) 1
- c) zero
- d) -2.5

17. By filling a lung with saline solution, what change occurs in the elastic forces of the lung compared to an air-filled lung?

- a) The elastic forces increase.**
- b) The elastic forces do not change.
- c) Only the elastic force due to elastin and collagen is preserved and functional.
- d) The distending pressures required to overcome the elastic forces increase.

18. What effect does alkalosis have on the oxyhemoglobin dissociation curve?

- a) It decreases hemoglobin's affinity for oxygen, causing a shift to the right.
- b) It increases hemoglobin's affinity for oxygen, causing a shift to the left.
- c) It decreases hemoglobin's affinity for oxygen, causing a shift to the left.
- d) It increases hemoglobin's affinity for oxygen, causing a shift to the right.

19. Clinically, the measurement of pulmonary wedge pressure is performed to estimate the pressure in which part?

- a) Right atrial pressure
- b) Pulmonary artery systolic pressure
- c) Pulmonary artery diastolic pressure
- d) Left atrial pressure

20. In the dynamics of fluid exchange across pulmonary capillaries in a healthy person, which Starling force has the greatest contribution towards the exit of fluid from the capillary to the interstitial space?

- a) Colloid osmotic pressure of the interstitial fluid
- b) Capillary hydrostatic pressure
- c) Negative pressure of the interstitial space
- d) Plasma colloid osmotic pressure

21. In a healthy, standing person, the value of which factor is lower at the base of the lung than at its apex?

- a) Alveolar ventilation rate
- b) Blood flow rate
- c) Ventilation to perfusion ratio (V/Q)
- d) Capillary hydrostatic pressure

22. If the ventilation/perfusion ratio becomes zero, what will happen to the partial pressure of oxygen and carbon dioxide in the alveoli?

- a) The partial pressure of oxygen will reach 149 mmHg.
- b) The partial pressure of carbon dioxide will reach 45 mmHg.
- c) The partial pressure of oxygen will become zero.
- d) The partial pressure of carbon dioxide will become zero.

23. What effect does the weakening of the pneumotaxic center in the pons have on the breathing pattern?

- a) The respiratory rate increases.
- b) The apneustic center becomes stronger.
- c) Breathing stops.
- d) The duration of the ramp signal produced in the DRG increases.

24. Where is the partial pressure of oxygen the highest?

- a) Arterial blood
- b) Alveolar air
- c) Expired air
- d) Anatomic dead space

25. Which of the following factors causes a decrease in lung compliance?

- a) Increased accumulation of elastin and collagen in the lung parenchyma.
- b) Increased amount of surfactant in the alveoli.
- c) Emphysema
- d) Sympathetic stimulation

26. Which of the following factors does not change during exercise?

- a) Pulmonary blood flow rate
- b) Pulmonary ventilation rate
- c) Partial pressure of oxygen in alveolar air
- d) Blood flow velocity through pulmonary capillaries

27. If in a patient with anemia the amount of hemoglobin in the blood is 10 g/dL and its oxygen saturation is 90%, approximately how many milliliters of oxygen will be present per deciliter of this person's blood, bound to hemoglobin?

- a) 10
- b) 12
- c) 14
- d) 16

28. Which process does NOT occur during the stimulation of carotid chemoreceptors due to a decrease in oxygen pressure?

- a) Entry of calcium into the glomus cell
- b) Release of acetylcholine from the glomus cell
- c) Depolarization
- d) Increased potassium efflux from the glomus cell

29. The magnitude of the safety factor against the development of pulmonary edema is determined by which factors?

- a) Plasma colloid osmotic pressure and capillary hydrostatic pressure
- b) Interstitial fluid colloid osmotic pressure and capillary hydrostatic pressure
- c) Plasma colloid osmotic pressure and interstitial fluid colloid osmotic pressure
- d) Plasma colloid osmotic pressure and negative pressure of the interstitial space